PH3105: Nuclear Physics Laboratory

Experiment 2

Ayush Chandra [*] 22MS107 | Susnata Chattopadhyay [†] 22MS077 | Niravra Nag [‡] 22MS072

Indian Institute of Science Education and Research, Kolkata,
Mohanpur, West Bengal, 741246, India

November 29, 2024

1 Aim

To study the absorption of γ-ray in matter (lead block) by a $^{60}_{27}$Co source. Then verify the inverse-square-law relationship of γ-ray radiation.

2 Experimental Setup

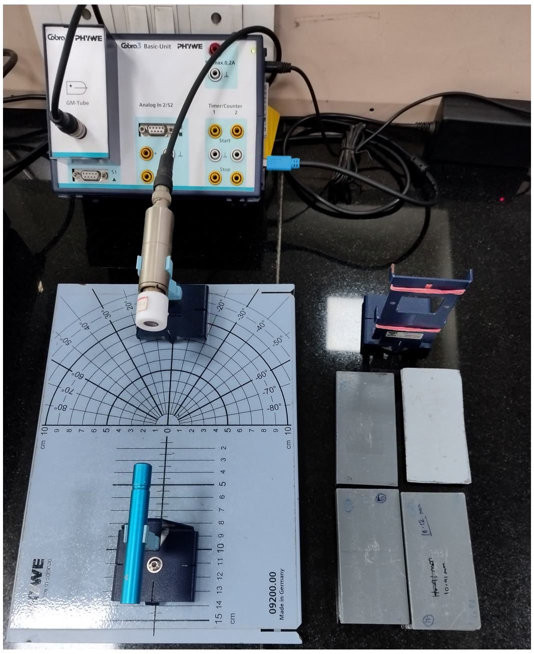

Figure 1: Experimental setup of the instruments in the laboratory

The experimental setup, shown in Figure 1, was designed to achieve the two primary objectives of the experiment.

The setup consists of a baseboard marked with distance and angular scales to facilitate precise positioning of the radiation source and detector. The source and detector are secured to magnetic holders placed on the board. The detector is connected to a Geiger-Müller (GM) counter for measuring radiation counts.

[*] ac22ms107@iiserkol.ac.in [†] sc22ms077@iiserkol.ac.in [‡] nn22ms072@iiserkol.ac.in

Determination of Attenuation Parameters of Pb Blocks

For this part of the experiment:

1. The radiation source and detector were fixed at specific positions on the baseboard and aligned so that they directly faced each other.
2. A holder for the lead (Pb) blocks was placed between the source and detector.
3. Pb blocks of varying thicknesses were inserted into the holder, and the corresponding counts were recorded on the GM counter.

This process allowed for determining attenuation parameters by analyzing how the radiation intensity decreased with increasing block thickness.

Verification of the Inverse Square Law

To verify the inverse square law:

1. The Pb blocks and their holder were removed from the setup.
2. The source and detector were positioned close to each other, with their relative distance measured from a reference zero point on the baseboard.
3. The position of the source holder was recorded, and the GM counter measured the radiation count.
4. The source was gradually moved farther from the detector, and counts were taken at multiple distances.
5. The recorded counts were used to plot a graph of count rate versus the inverse square of the distance.

3 Working Principle

Geiger Müller (GM) Counter

These detectors work by using an electric field to collect and count ion pairs (electrons and positive ions) created when radiation ionizes a gas inside a sealed chamber. The GM counter consists of a cylindrical metallic chamber that serves as the cathode, with a fine wire along its axis as the anode (depicted in figure below). Inert (noble) gases are chosen to fill the counter for their low electron affinity - they do not tend to form negative ions. Argon is most common fill gas: It is inert, stable (non-radioactive), abundant,inexpensive, has reasonably high atomic number and works well with halogen quenching gases (See Fig 2).

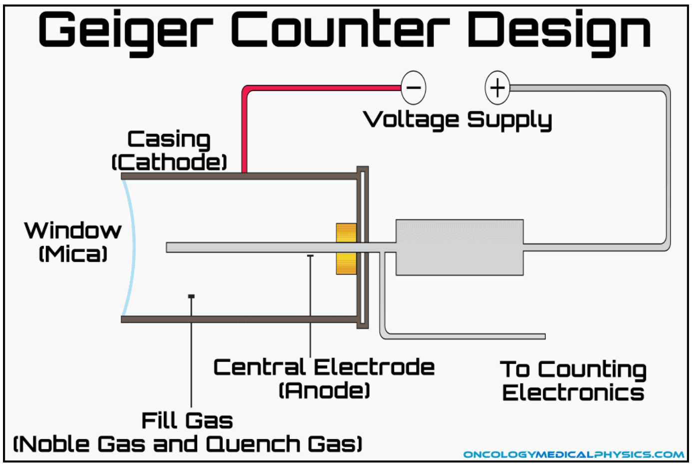

Figure 2: Schematic of GM counter

In a cylindrical geometry, electric field($\mathcal{E}$) at a radial distance r from the center is given by

$$\mathcal{E}(r) = \frac{V}{r \ln\left(\frac{b}{a}\right)}$$

where V is voltage applied between anode and cathode, a is anode wire radius, b is cathode inner radius.

This shows that the electric field strength for a given voltage is much larger in cylindrical geometry than in a parallel plate one, which makes the experimental conditions easier to attain. Another advantage of the cylindrical geometry is that *avalanche* (amplification) takes place only in narrow region close to the anode wire which gives good uniformity.

Townsend Avalanche

The chamber has a sealed window through which radiation like alpha (α), beta (β), or gamma (γ) particles enter and ionize the gas, typically argon at low pressure.Usually monatomic gas is preferred so that deposited energy is not dissipated in dissociating the molecules. This ionization produces free electrons and positive ions. A high voltage applied across the chamber generates an electric field that accelerates these free electrons toward the anode and the positive ions toward the cathode. This movement of ions creates a current pulse as they reach their respective electrodes. To amplify the initial ionization, the GM counter operates at a high voltage (often over 1000 V), allowing the electrons to gain enough energy to cause secondary ionisations as they collide with other gas atoms in a process called a *Townsend Avalanche*. This avalanche effect exponentially increases the number of ions and electrons, resulting in a strong pulse of current that the counter registers (See Fig 3).

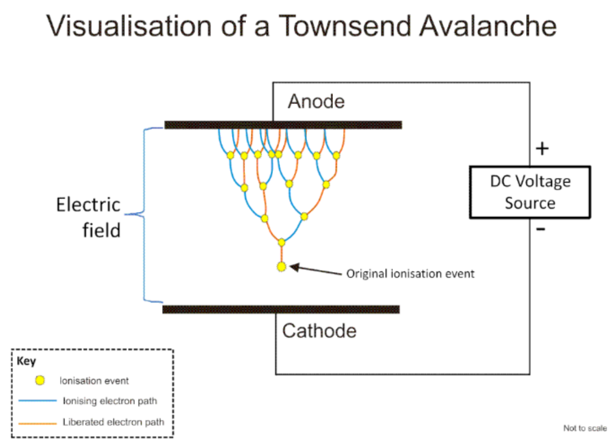

Figure 3: Townsend Avalanche. Source: Wikipedia

However, without control, the GM tube might experience continuous discharge, making it unresponsive to new ionizing events. To prevent this, a *quenching agent*, such as butane, propane, or halogen gases (like chlorine or bromine), is added. The quenching agent absorbs excess energy and neutralizes positive ions without triggering further ionization. In organic quenching gases, energy from the positively charged ions is dissipated by breaking molecular bonds, which stops continuous discharge. Once the ions reach the electrodes and are neutralized, the tube resets, ready to detect the next radiation event. If continuous ionization is not prevented the detector will not respond to any other particle except the very first which enters the tube. The GM counter's design makes it suitable for measuring low to moderate levels of radiation, though it cannot provide information about the energy of incoming particles since all pulse sizes are identical regardless of the intensity of the radiation. Due to its simplicity and robustness, the GM counter is popular as a portable radiation detector, though it is limited to applications requiring simple presence detection rather than detailed energy spectroscopy.

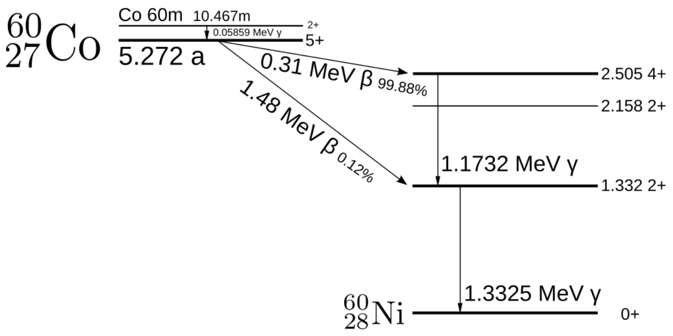

Figure 4: Decay scheme of our source. Source: Wikipedia

4 Theory

Decay of Co-60

$^{60}_{27}$Co decays into an excited $^{60}_{28}$Ni nucleus (Fig 4), producing a β^- and an antineutrino.

$$^{60}_{27}\mathrm{Co} \rightarrow {}^{60}_{28}\mathrm{Ni}^* + \beta^- + \bar{\nu}_e$$

Then the excited $^{60}_{28}$Ni nucleus then decays into ground state $^{60}_{28}$Ni nucleus in a 2 step process, producing two γ-rays of different energies.

$$^{60}_{28}\mathrm{Ni}^* \rightarrow {}^{60}_{28}\mathrm{Ni} + \gamma(1.1732 \text{ MeV}) + \gamma(1.3325 \text{ MeV})$$

This decay process has a half life of approximately 5.27 years.

Mass attenuation

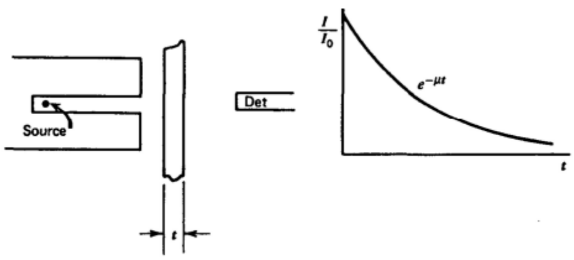

Figure 5: The exponential transmission curve for gamma ray

If we picture a transmission experiment as in the figure above, where monoenergetic gamma rays are collimated into a narrow beam and allowed to strike a detector after passing through an absorber of variable thickness, the result should be simple exponential attenuation of the gamma rays. Each of the interaction processes removes the gamma-ray photon from the beam either by absorption or by scattering away from the detector direction and can be characterized by a fixed probability of occurrence per unit path length in the absorber. The sum of these probabilities is simply the probability per unit path length that the gamma-ray photon is removed from the beam:

$$\mu = \tau(\text{photoelectric}) + \sigma(\text{Compton}) + \kappa(\text{pair})$$

This is called the *linear attenuation coefficient* (μ). Empirically, this is defined as

$$\boxed{\mu = \frac{-1}{I}\frac{\mathrm{d}I}{\mathrm{d}z}}$$

Thus integrating we have

$$\int_{I_0}^{I} \frac{\mathrm{d}I}{I} = -\int_0^t \mu \mathrm{d}z$$

,

$$\implies \log(I) - \log(I_0) = -\mu t$$
$$\implies \log\left(\frac{I}{I_0}\right) = -\mu t$$

Thus, the number of transmitted photons I (per thickness t of the absorber) is then given in terms of the number without an absorber I_0 as

$$\frac{I}{I_0} = e^{-\mu t}$$

Use of the linear attenuation coefficient is limited by the fact that it varies with the density of the absorber, even though the absorber material is the same. Therefore, the *mass attenuation coefficient* is much more widely used and is defined as

$$\kappa = \frac{\mu}{\rho}$$

where ρ represents the density of the medium. For a given gamma-ray energy, the mass attenuation coefficient does not change with the physical state of a given absorber. For example, it is the same for water whether it is present in liquid or vapor form.

Thus, the attenuation law for gamma rays in terms of mass attenuation coefficient takes the form

$$\boxed{\frac{I}{I_0} = e^{-\kappa\rho t}}$$

The product ρt, known as the *mass thickness* of the absorber, is now the significant parameter that determines its degree of attenuation. The thickness of the absorber used in radiation measurement is therefore often measured in mass thickness rather than physical thickness, because it is a more fundamental physical quantity.

Inverse Square Law

The inverse square law states that the intensity I of radiation decreases with the square of the distance r from the source, i.e.,

$$I(r) \propto \frac{1}{r^2}$$

where:

- $I(r)$ is the intensity of the radiation at distance r from the source,
- r is the radial distance from the point of emission.

5 Data Analysis

Background Reading

The apparatus was set up without any source, and counts were taken using a GM counter to determine the background reading. The average of two readings, each for 600 seconds, was calculated as follows:

$$\text{Count 1} = 173$$
$$\text{Count 2} = 153$$
$$\therefore \text{Average Background Count rate} \equiv I_{bg} = \frac{163}{600\text{ s}} = 0.272\text{ s}^{-1}$$

This background reading was subtracted from each of the main readings to obtain the corrected readings.

Measuring Thickness of Lead Blocks

1. Measurement Using Screw Gauge

For the screw gauge, we observed the following:

- 1 Main Scale Division (MSD) corresponds to 0.5 mm of length.
- 100 Circular Scale Divisions (CSD) correspond to 1 MSD.

$$\therefore 1 \text{ CSD} = \frac{1}{50} \times 1 \text{ MSD} = 0.01 \text{ mm}$$

$$\therefore \text{Least Count (LC)} = 0.01 \text{ mm}$$

The zero error was found to be 0.00 mm. The thin blocks were measured using this screw gauge, and the results are shown in Table 1.

Block No.	Main Scale Reading (mm)	Circular Scale Reading	Total Reading (mm)
1	1	7	1.07
2	1	12	1.12
3	1	8	1.08
4	1	3	1.03
5	1	5	1.05

Table 1: Measurements Using Screw Gauge

Measurement Using Vernier Calipers

For the vernier calipers, the following observations were made:

- 1 Main Scale Division (MSD) corresponds to 1 mm of length.
- 50 Vernier Scale Divisions (VSD) correspond to 49 MSD.

$$\therefore 1 \text{ VSD} = \frac{49}{50} \times 1 \text{ MSD} = 0.98 \text{ mm}$$

$$\therefore \text{Least Count (LC)} = 1 \text{ MSD} - 1 \text{ VSD} = 1 \text{ mm} - 0.98 \text{ mm} = 0.02 \text{ mm}$$

The zero error was found to be 0.00 mm. The thick blocks were measured using this vernier calipers, and the results are shown in Table 2.

Block No.	Main Scale Reading (mm)	Vernier Scale Reading	Total Reading (mm)
6	5	27	5.54
7	10	6	10.12
8	10	7	10.14
9	10	2	10.04
10	10	0	10.00

Table 2: Measurements Using Vernier Calipers

Part - 1: Determining Mass Attenuation Coefficient

We define the mass attenuation coefficient (normalised per density) as $\kappa = \mu/\rho$. For our experiment, the density of lead (Pb) blocks are $\rho = 1.1343 \times 10^4$ kg/m^3 = 1.1343×10^{-2} g/mm^3. We have,

$$\frac{I}{I_0} = \exp\{-\kappa\rho t\}$$
$$\log I = \log I_0 - \kappa\rho t$$

Here I_0 refers to the count rate (counts/s) observed in the absence of any obstruction ($t = 0$). We perform a linear fit ($y = a_0 + a_1 x$) such that the parameters are

$$a_0 = \log I_0$$
$$a_1 = -\kappa\rho \implies \kappa = -a_1/\rho$$

NOTE: We plot the logarithm of the *corrected count rate* ($I = I_{obs} - I_{bg}$) with the thickness (t) of the blocks in mm. (Data is in Table 3). The results of the (Least squares) Linear fit are summarized below:

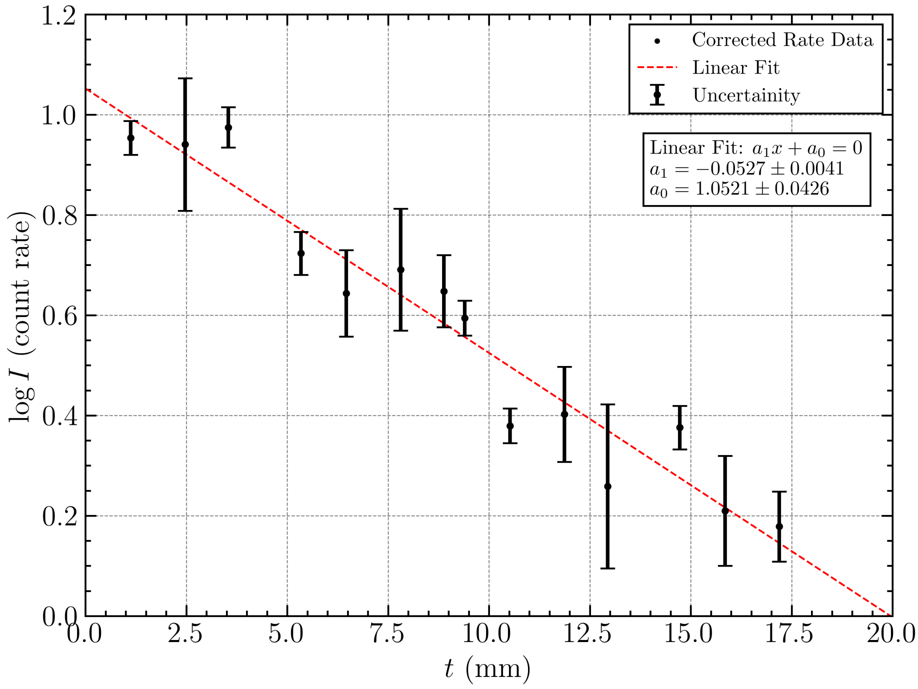

Figure 6: Linear Fit of the corrected (background) count rates with varying lead block thicknesses. The uncertainties in y-values are reported as $\Delta(\log I) = \frac{\sigma_I}{I}$.

$$a_1 = -0.0527 \pm 0.0041$$

$$a_0 = 1.0521 \pm 0.0426$$

Hence the mass attenuation coefficient (κ) of lead block corresponding to Co-60 γ-ray source is calculated to be

$$\kappa = -a_1/\rho = 4.6493 \text{ mm}^2/\text{g}$$

Calculation of half value thickness

Half value thickness (t_h) is defined as the thickness of the absorber at which the value of the incident intensity is attenuated to half of its initial value. We have,

$$I_0/2 = I_0 \exp\{-\kappa\rho t_h\} \implies \boxed{t_h = \frac{\ln 2}{\kappa\rho}}$$

Calculating we get $t_h = 13.1434$ mm.

Part - 2: Verifying Inverse Square Law

For the second part of the experiment, we vary the distance between the source and the detector and measure the count rates. We plot this count rate I (background corrected) versus distance of source and fit it with function $f_{inv}(x)$, which we shall define soon. By, the inverse square law, we have the incident count rates I to vary as

$$I = \frac{I_0'}{(x + l_o)^2}$$

where x is the (apparent) distance between source and detector (in cm). Since the distance measurements are not accurate enough we have added an offset l_o, which we shall determine through fitting. I_0' is a constant of appropriate dimensions. Note that, by taking logarithm on both sides, the above expression can be re-written as

$$I = \exp\Big\{\ln I_0' - 2\ln(x + l_o)\Big\}$$

$$\downarrow$$

Consequently we define f_{inv} to fit our data points

$$\boxed{f_{inv}(x) \equiv e^{(a+b\ln(x+c))}}$$

where we identify $a \sim \ln I_0'$, $c \sim l_0$ (cm) and expect b to be close to -2 for a successful validation of this law. The results of the fitting are summarized below

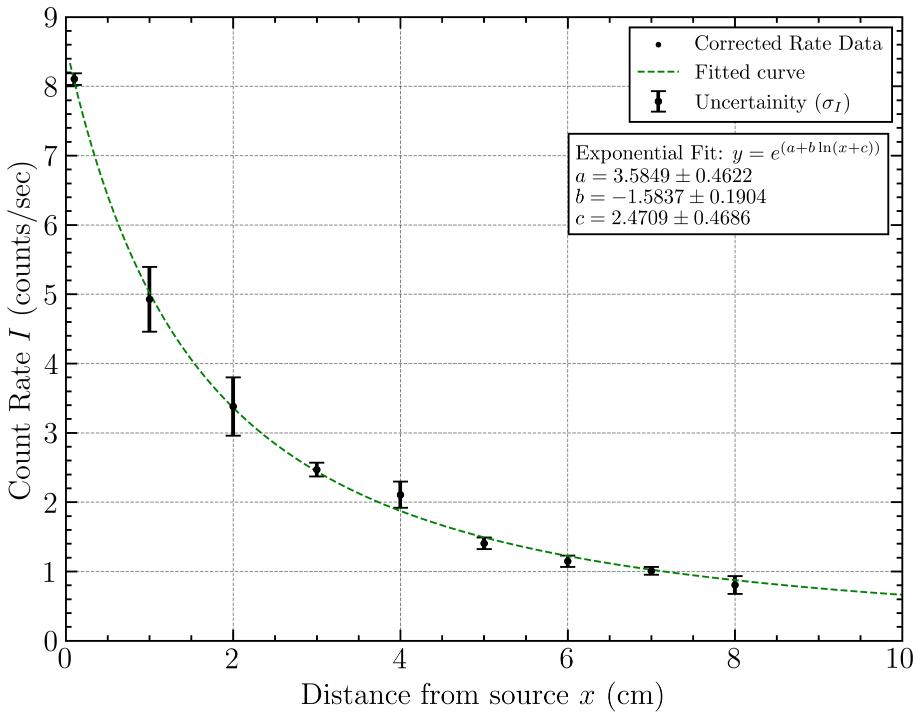

Figure 7: Graph to validate the inverse square law relationship. The uncertainty in measuring I is just the standard deviation (σ_I) of all the count rates

$$a = 3.5849 \pm 0.4622$$

$$b = -1.5837 \pm 0.1904$$

$$c = 2.4709 \pm 0.4686$$

From the fit, we obtain the value of b to be close -2, thus **successfully validating** the inverse square law relationship. The reason for this deviation is explained in the sources of errors section.

6 Error Analysis

Error in κ

Note that, $\kappa = -a_1/\rho$. Thus maximizing error $|\Delta\kappa| = |\Delta a_1|/\rho = 0.3628$. Where Δa_1 denotes the fitting error in slope. Thus we measure $\boxed{\kappa = 4.6493 \pm 0.3628 \text{ mm}^2/\text{g}}$

Error in t_h

Note that, $t_h = \frac{\ln 2}{\kappa\rho}$. Thus maximizing error $|\Delta t_h| \leq \frac{|\Delta\kappa| t_h}{\kappa}$. Thus we measure $\boxed{t_h = 13.1434 \pm 1.0255 \text{ mm}}$

Error in Validating Inverse Square Law

In our results, $b = -1.5837$, while the expected value is -2. Hence, the percentage error is $\frac{|b-(-2)|}{|-2|} \times 100 \approx$ 21%.

Sources of Error

Environmental Interference: The presence of other ongoing experiments in the same room likely caused radiation leakage, which could have contributed to false readings in our setup. This external interference may have led to an overestimation of the count rate.

Alignment of GM Counter The GM counter tube needed to be perfectly aligned with the source emission otherwise there would be an angle which would cause deviation from expected result. In our experiment this was done with an eyeball estimate.

Error in Alignment of Plates The plates were not all perfectly parallel and this caused inaccurate measurement of combined thickness. The air gaps between plates might have also led to surface scattering effects, in part A.

Error in Distance Calculation The inverse square law holds for distance between the source point of emission and the point of detection. In the experiment, the exact point of detection was not known, thus giving us an offset for the distance measured. In our experiment we try to reduce this error by fitting with an offset parameter which is determined to be $l_o = 2.4709 \pm 0.4686$ cm.

Deviation of inverse square law Due to the absorption and scattering of radiations by air molecules, inverse square law deviates slightly from ideal behavior. Error associated with isotropy also contributed in the deviation of inverse square law.

7 Conclusion

The mass attenuation coefficient (κ) of lead block corresponding to Co-60 γ-ray source was measured, and is found to be $\boxed{\kappa = 4.6493 \pm 0.3628 \text{ mm}^2/\text{g}}$. The half value thickness of lead block was also found out to be $\boxed{t_h = 13.1434 \pm 1.0255 \text{ mm}}$ We also experimentally verified the inverse square law, within a marginal degree of error. The distance offset between the source and detector (in Part-B) was also calculated and found to be $l_o = 2.47 \pm 0.47$ cm.

Appendix

Time (s)	Thickness (mm)	Count	Individual Count Rates	Average Counts	I_{obs} (counts/s)	σ_I
30	1.12	89	2.967	86	2.867	0.0882
		85	2.833			
		84	2.800			
30	2.47	76	2.533	85	2.833	0.3383
		96	3.200			
		83	2.767			
30	3.54	84	2.800	88	2.922	0.1072
		90	3.000			
		89	2.967			
30	5.34	67	2.233	70	2.333	0.0882
		71	2.367			
		72	2.400			
40	6.46	93	2.325	87	2.175	0.1639
		80	2.000			
		88	2.200			
40	7.81	99	2.475	91	2.267	0.2428
		93	2.325			
		80	2.000			
40	8.88	87	2.175	87	2.183	0.1377
		82	2.050			
		93	2.325			
40	9.40	86	2.150	83	2.083	0.0629
		83	2.075			
		81	2.025			
50	10.52	84	1.680	87	1.733	0.0503
		87	1.740			
		89	1.780			
50	11.87	82	1.640	88	1.767	0.1419
		96	1.920			
		87	1.740			
50	12.94	67	1.340	78	1.567	0.2120
		80	1.600			
		88	1.760			
60	14.73	101	1.683	104	1.728	0.0631
		102	1.700			
		108	1.800			
70	15.85	98	1.400	105	1.505	0.1350
		102	1.457			
		116	1.657			
70	17.19	105	1.500	103	1.467	0.0837
		107	1.529			
		96	1.371			

Table 3: Data for Part-A of the experiment.

Distance of Source (cm)	Time (s)	Count	Individual Count Rates	I_{obs} (counts/s)	σ_I
0.1	30	251	8.367	8.106	0.0839
		254	8.467		
		249	8.300		
1	30	162	5.400	4.928	0.4667
		140	4.667		
		166	5.533		
2	50	190	3.800	3.381	0.4197
		159	3.180		
		199	3.980		
3	70	200	2.857	2.471	0.1000
		187	2.671		
		189	2.700		
4	70	158	2.257	2.109	0.1902
		160	2.286		
		182	2.600		
5	80	127	1.588	1.407	0.0832
		140	1.750		
		136	1.700		
6	100	135	1.350	1.148	0.0819
		140	1.400		
		151	1.510		
7	150	199	1.327	1.01	0.0555
		183	1.220		
		195	1.300		
8	150	159	1.060	0.806	0.1276
		182	1.213		
		144	0.960		

Table 4: Data for Part- B of the experiment

References

- Knoll, Glenn F. *Radiation detection and measurement.* John Wiley & Sons, 2010.